

Assignment 2

Database Schema Documentation
Chicago Weather Trend Analytics Project

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Data Source: Open-Meteo REST API

1. Project Information

Database name: chicago_weather_db
RDBMS: PostgreSQL 15
Total tables: 1
City: Chicago, Illinois
Coordinates: Latitude 41.85 N, Longitude -87.65 W
Elevation: 179 m above sea level
Timezone: America/Chicago (GMT-5)
Data source: Open-Meteo REST API (<https://open-meteo.com>)
Units: Temperature in Fahrenheit, Precipitation in inches

2. Table Definition

2.1 Table name: weather_data

This table stores daily weather records for Chicago, Illinois. Each row represents one calendar day and may contain either a historical observation or a future forecast value, as both are returned together by the Open-Meteo API response.

Column name	Data type	Constraint	Nullable	Description
weather_id	SERIAL	PRIMARY KEY	NOT NULL	Auto-incrementing unique identifier for each row
forecast_date	DATE		NOT NULL	Calendar date of the weather record (YYYY-MM-DD)
temperature_max	DECIMAL(5,2)		NULL	Daily maximum temperature in degrees Fahrenheit
temperature_min	DECIMAL(5,2)		NULL	Daily minimum temperature in degrees Fahrenheit
precipitation_probability	DECIMAL(5,2)		NULL	Max precipitation probability for the day, as a percentage (0-100)
precipitation_total	DECIMAL(6,2)		NULL	Total precipitation accumulated for the day, in inches

3. Primary Key

Column: weather_id

Type: SERIAL PRIMARY KEY

The weather_id column is an auto-incrementing integer generated automatically by PostgreSQL each time a new row is inserted. It uniquely identifies every weather record in the table. No value needs to be provided manually during an INSERT statement.

4. Data Types Explained

SERIAL — PostgreSQL integer type that auto-increments by 1 with each new row inserted. Used for the primary key.

DATE — Stores a calendar date in YYYY-MM-DD format with no time component. Used for forecast_date.

DECIMAL(5,2) — Exact numeric type with up to 5 total digits and 2 decimal places (e.g. 999.99). Used for temperature columns.

DECIMAL(6,2) — Exact numeric type with up to 6 total digits and 2 decimal places (e.g. 9999.99). Used for precipitation_total.

NOT NULL — The column must always have a value. Applied to weather_id and forecast_date.

NULL — The column may be empty if the API does not return a value for that day.

5. API Field Mapping

The table columns map directly to the following fields returned in the Open-Meteo daily API response:

Table column	Open-Meteo API field	Notes
temperature_max	temperature_2m_max	Max temp at 2 metres above ground
temperature_min	temperature_2m_min	Min temp at 2 metres above ground
precipitation_probability	precipitation_probability_max	Highest probability value across the day
precipitation_total	precipitation_sum	Total precipitation summed over the day

6. Design Decisions

Single table design. Since this project is scoped to one city (Chicago), a single flat table is sufficient. There is no need for separate location or forecast tables.

Historical and forecast in one table. The Open-Meteo API returns both past observations and future forecast dates in the same JSON response payload. A single table naturally reflects this structure.

Nullable weather columns. Temperature and precipitation columns are set to NULL to handle cases where the API does not return a value for a given day, avoiding insert failures during the ETL pipeline run.

Units preserved as-is. All values are stored in the units returned by the API — Fahrenheit for temperature and inches for precipitation — which match the preferences of the target audience (Chicago residents and tourists).